

# Intelligent Product Discovery & Supply Chain Platform

Principal AI Architect Architecture Case Study

Engineering Knowledge Fabric • GraphRAG • Agentic AI • Supply Chain Intelligence • Regulatory Automation

JYOTIRMOY BARDHAN

# Executive Architecture Mandate

The architectural mandate was to replace disconnected AI workstreams with a reusable enterprise intelligence fabric. The platform reasons over relationships between products, manufacturers, substances, regulations, supply signals and engineer intent rather than relying on document similarity alone.

**Scale:** 1M+ SKUs • 2,000+ manufacturer relationships • 890K engineer community

**Five pillars:** Engineering BOM Intelligence • Supply Chain Resilience • Autonomous Content Factory • Regulatory Compliance Command Center • Community-to-Commerce Conversion

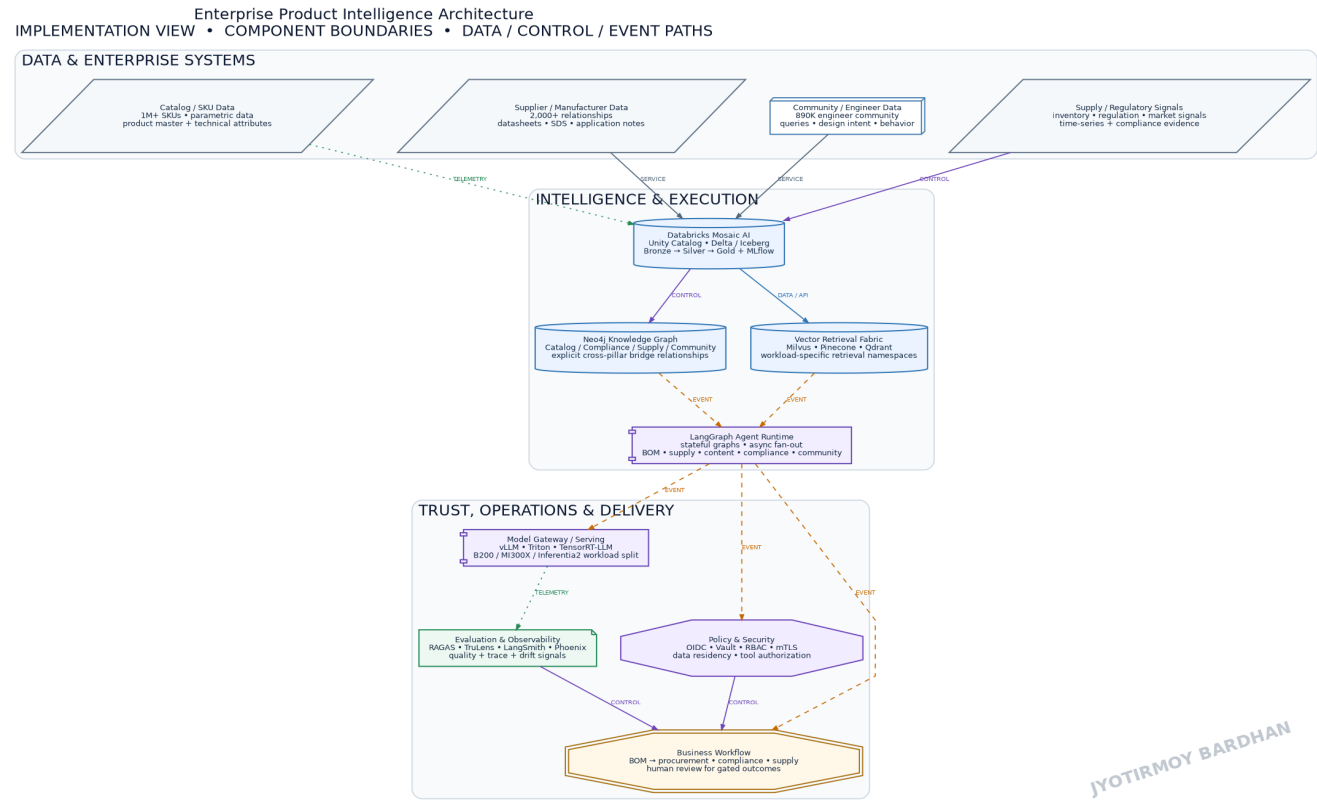
**Architecture position:** Neo4j provides relational truth; vector stores provide workload-specific semantic retrieval; LangGraph coordinates stateful workflows; Kafka/MSK propagates state changes; model routing and inference controls enforce quality, latency and cost constraints.

Model output does not become an authorization boundary. Compliance-sensitive outcomes are evaluation-gated and can be routed to human review.

# Architecture Evidence Map

| Requirement            | Decision                       | Implementation                                        | Documented evidence                   |
|------------------------|--------------------------------|-------------------------------------------------------|---------------------------------------|
| BOM latency            | Parallel stateful execution    | LangGraph async fan-out + cache + optimized retrieval | 2.1s P95 for 40-line BOM              |
| Relationship reasoning | Graph as relational truth      | Neo4j domain subgraphs + bridge relationships         | 18ms P95 compliance path              |
| Retrieval scale        | Workload-specific vector tiers | Milvus + Pinecone + Qdrant                            | 1M+ SKU semantic index                |
| Compliance safety      | Evaluation-gated publication   | RAGAS + TruLens + human review                        | 4.2% → 0.08% hallucination            |
| Inference economics    | Quality-per-dollar routing     | 8B / 70B / frontier routing + quantization            | 60% cost reduction                    |
| State consistency      | Event-driven reconciliation    | Kafka/MSK + cache + checkpoint validation             | zero contradictory outputs documented |
| Sovereignty            | Residency-aware placement      | GCP / EU / GovCloud / edge                            | ITAR/EAR air-gapped execution         |

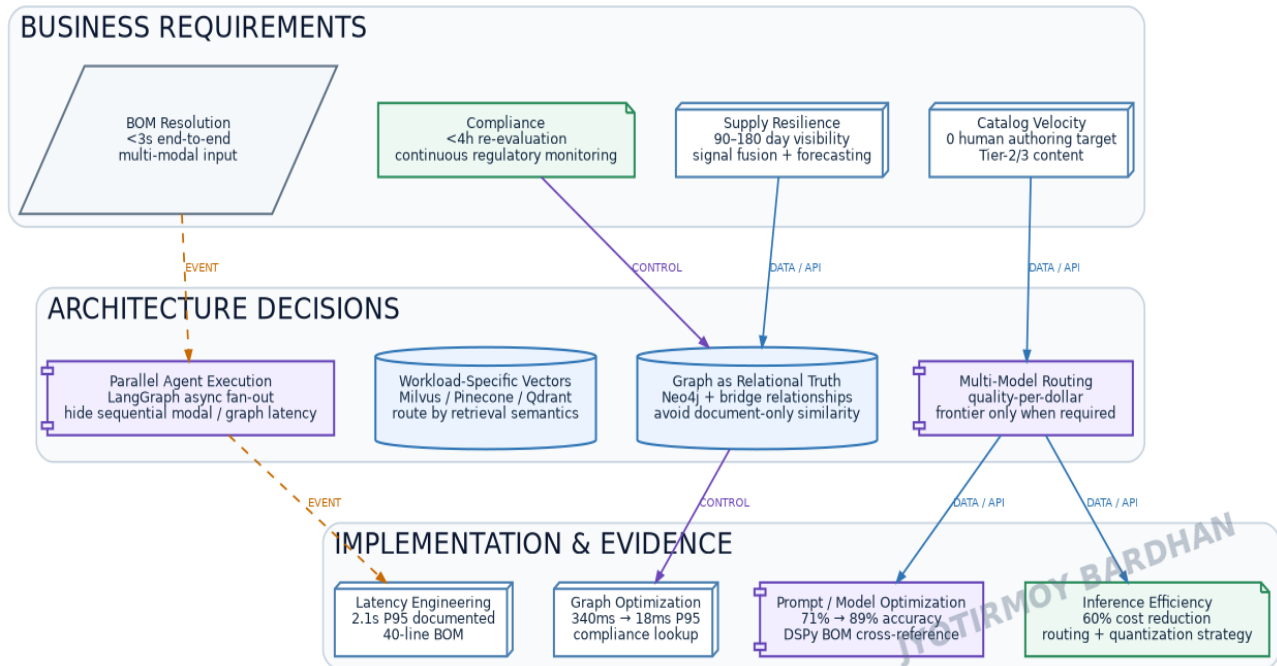
# 01. Enterprise Product Intelligence Architecture



Implementation view with typed architectural components, explicit data/API/event/control/telemetry paths, technology-specific boundaries and production evidence where documented.

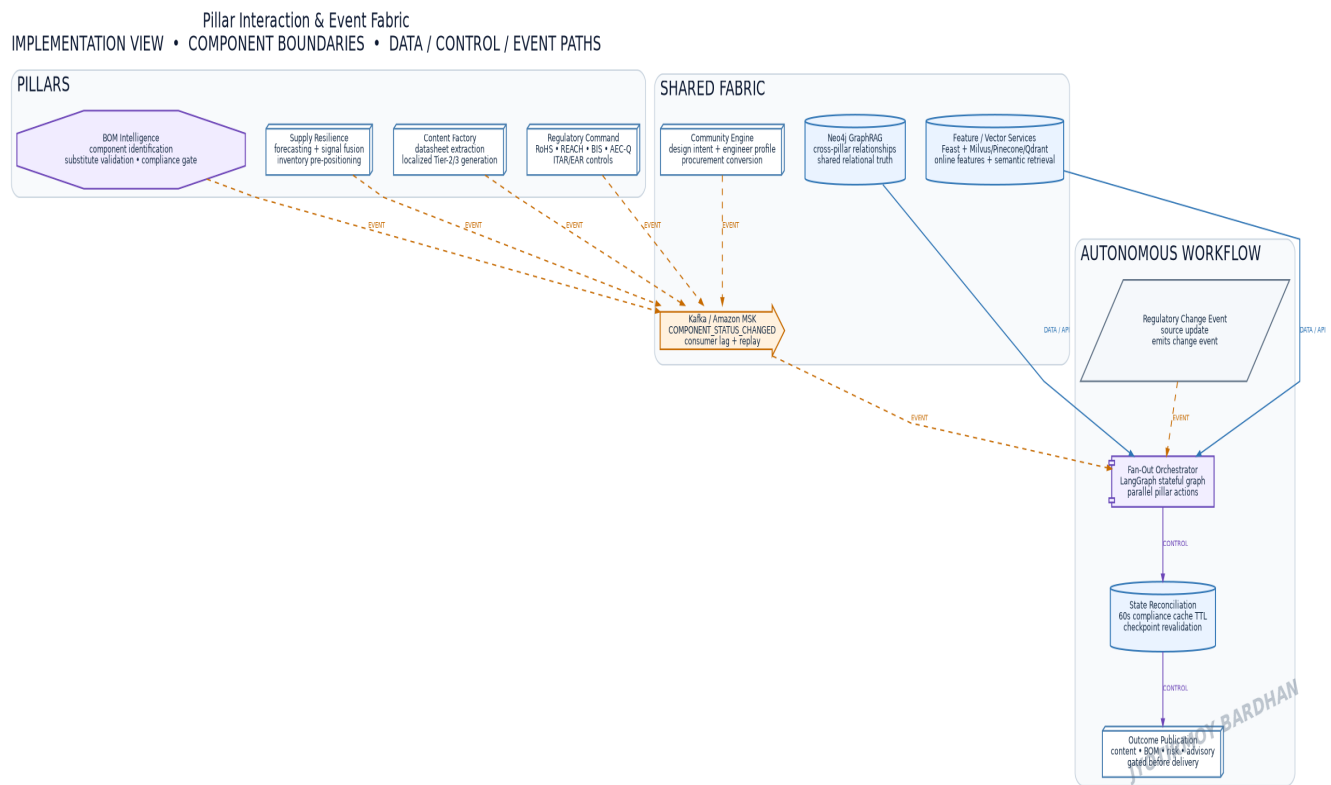
## 02. Architecture Evidence Map

Architecture Evidence Map  
IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS



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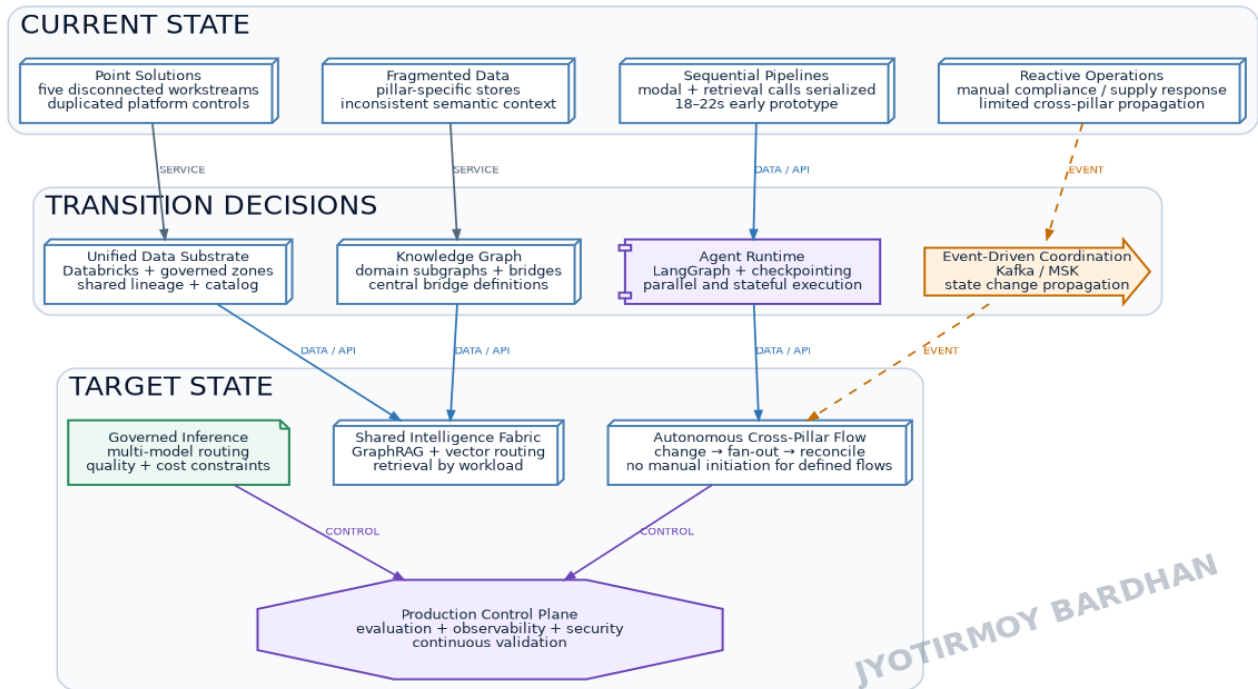
# 03. Pillar Interaction & Event Fabric



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## 04. Current-State → Target-State Transformation

Current-State → Target-State Transformation  
IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS

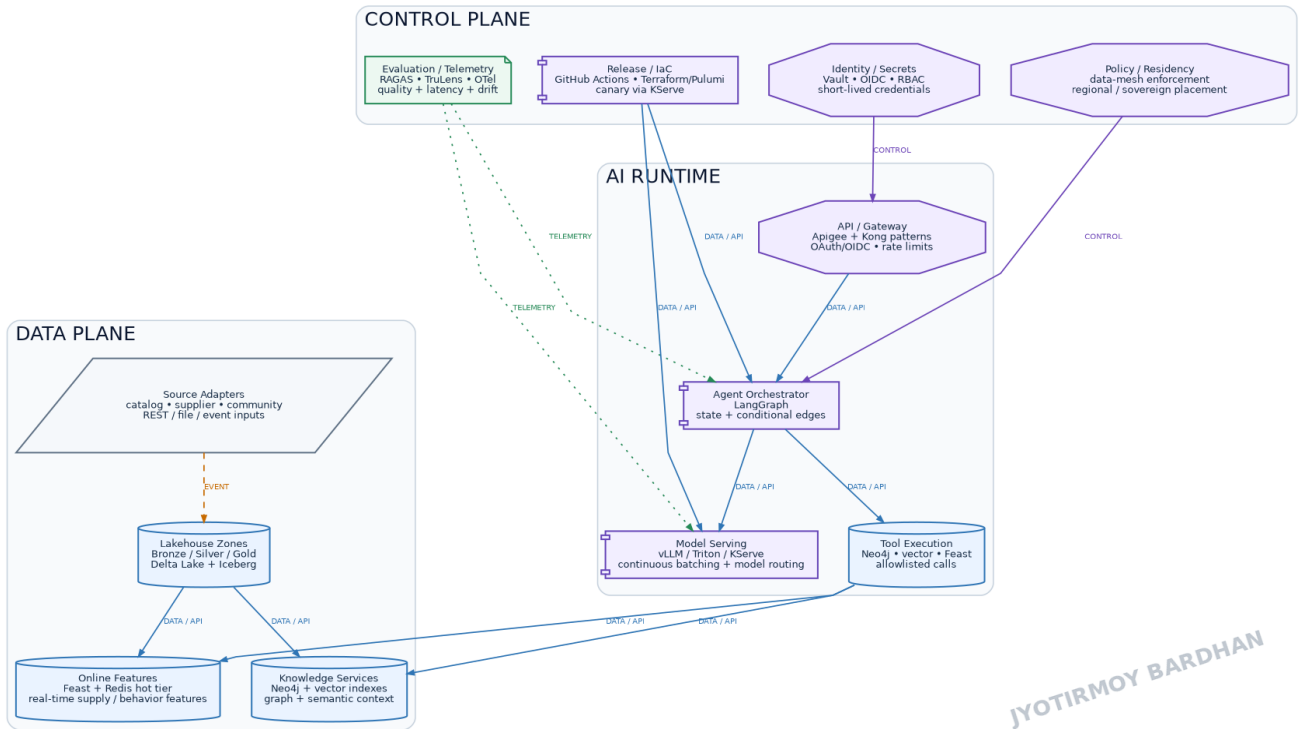


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# 05. High-Level Design — Data & Control Planes

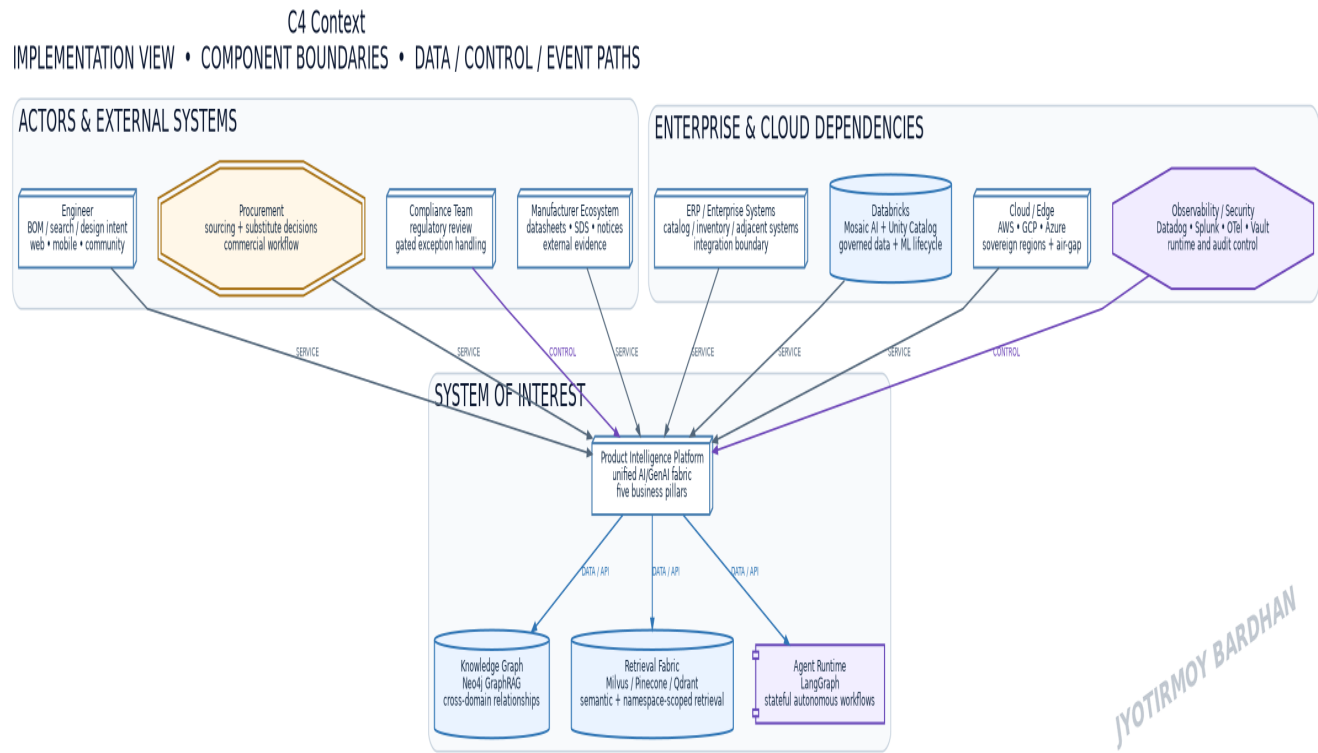
## High-Level Design — Data & Control Planes

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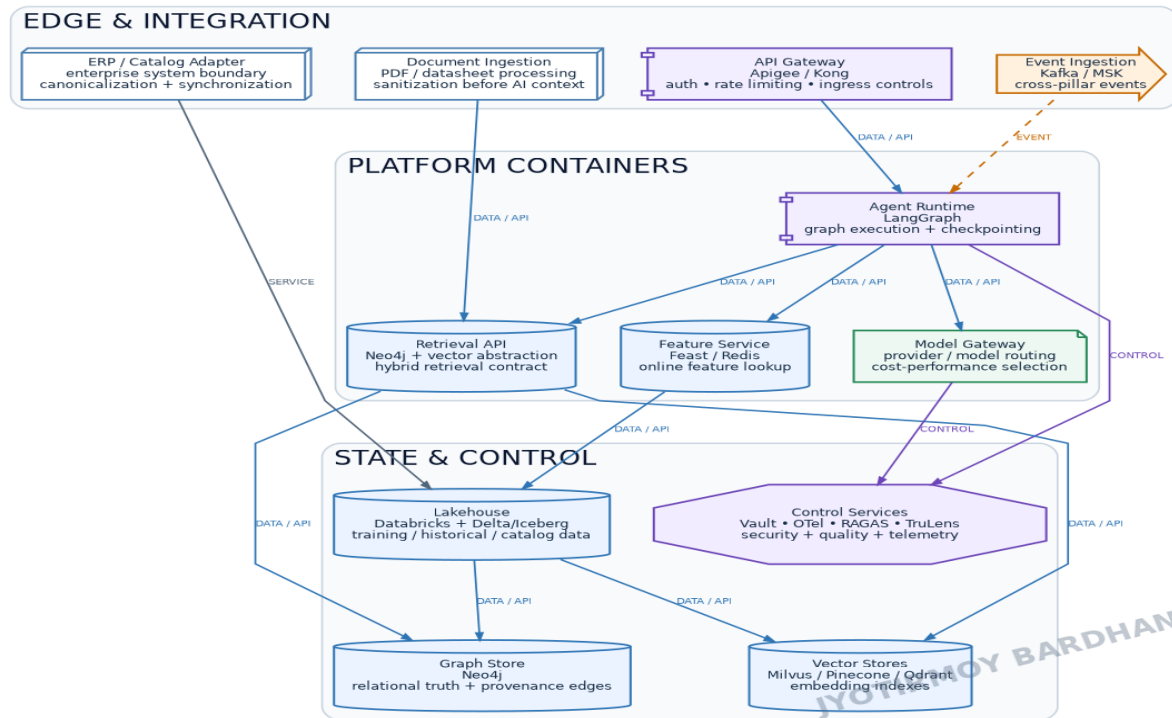
# 06. C4 Context



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## 07. C4 Container

C4 Container  
IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS

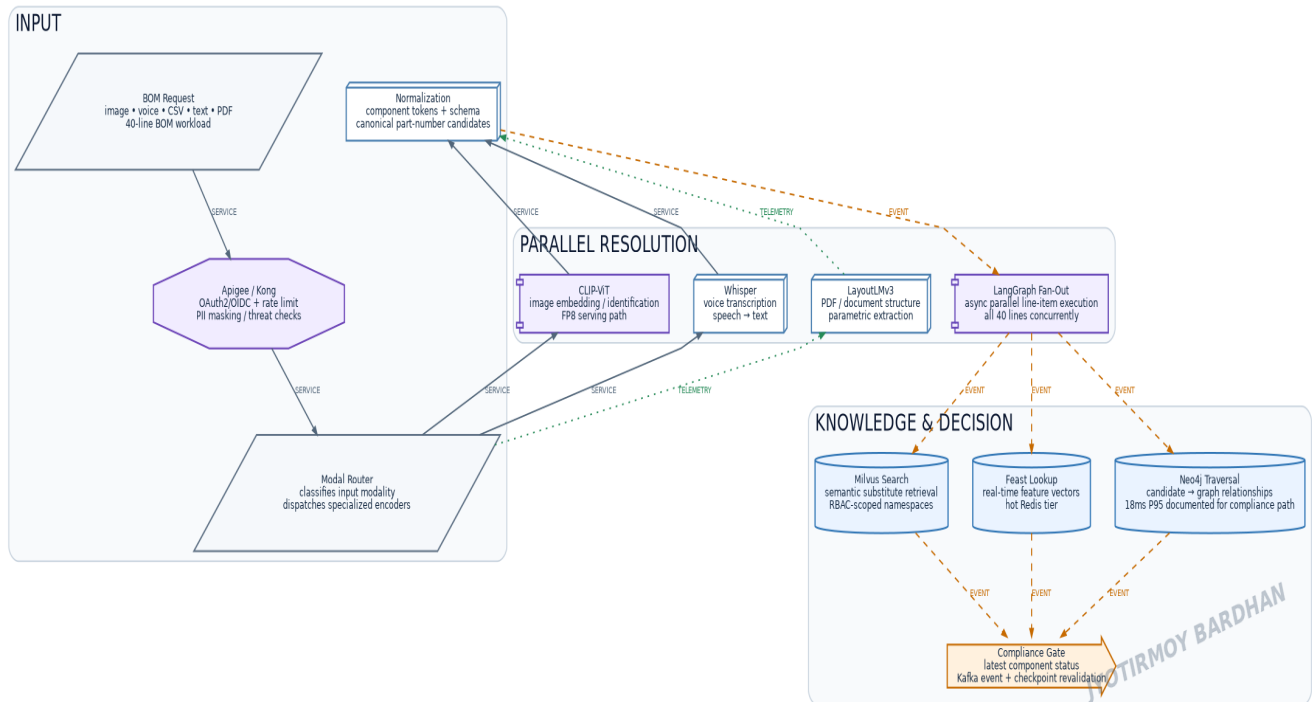


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## 08. LLD — BOM Multi-Modal Ingestion

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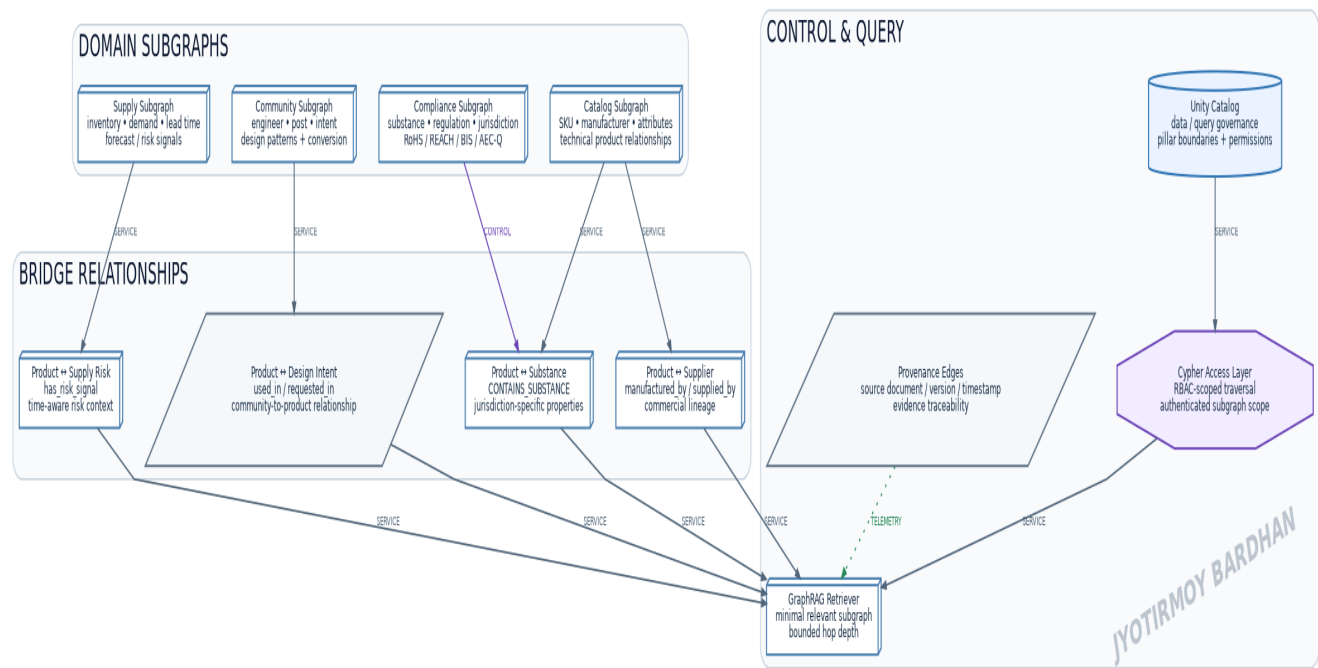
IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS



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# 09. LLD — Graph Knowledge Fabric

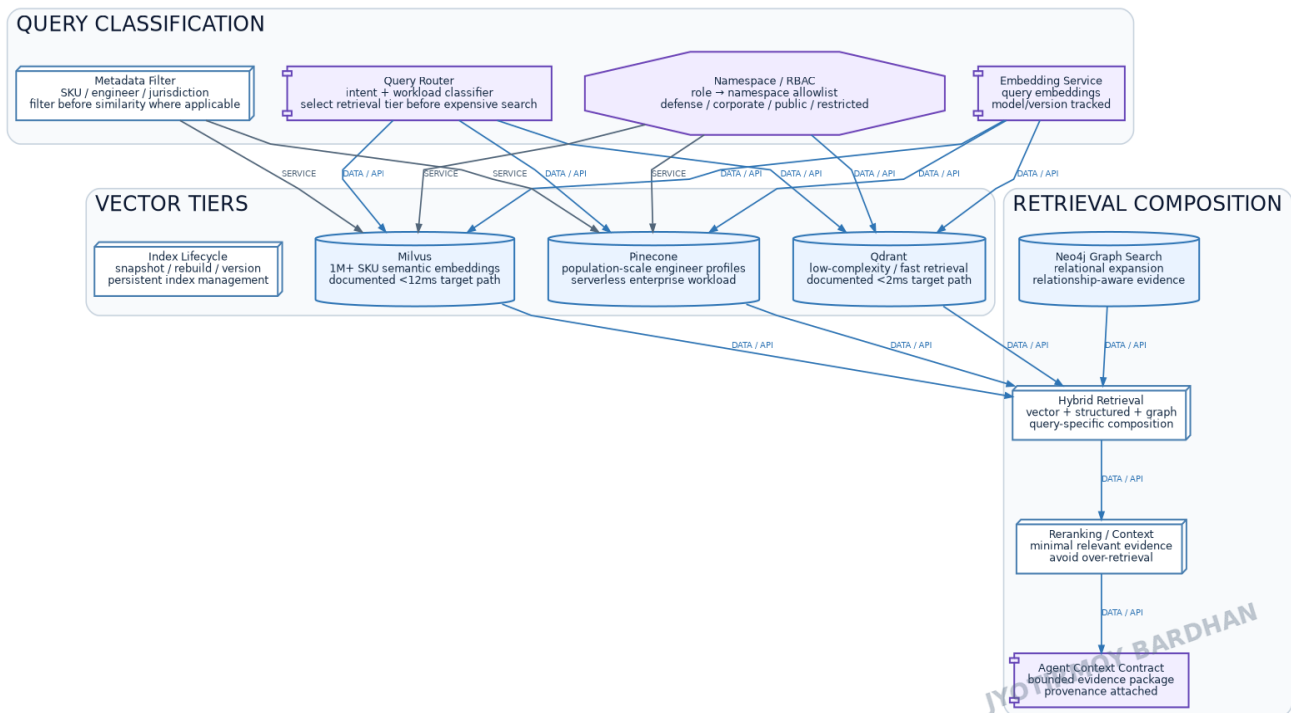
LLD — Graph Knowledge Fabric  
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# 10. Three-Tier Vector Architecture

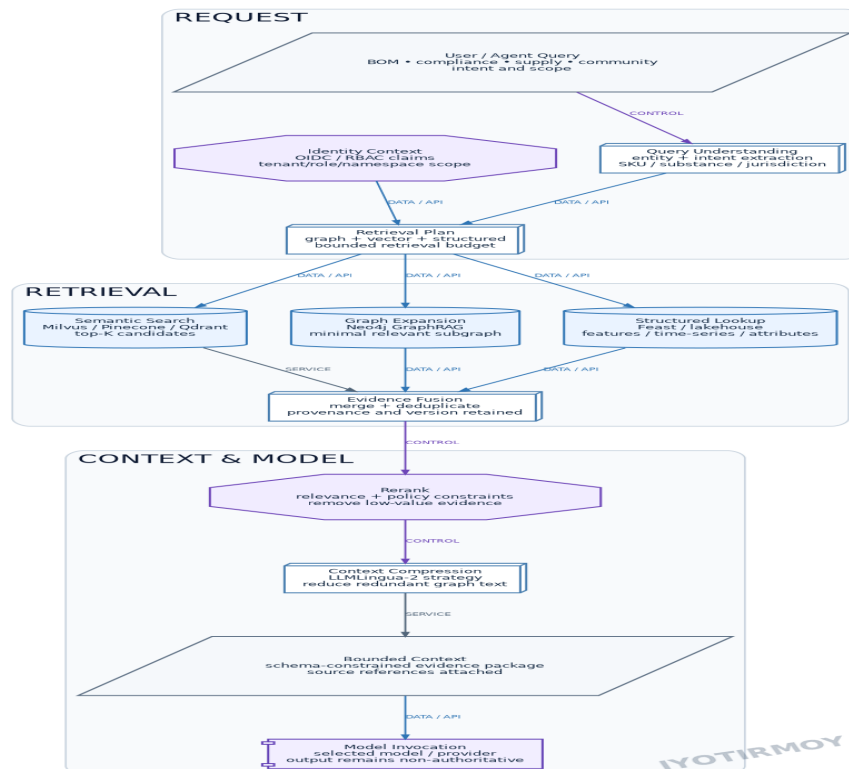
Three-Tier Vector Architecture  
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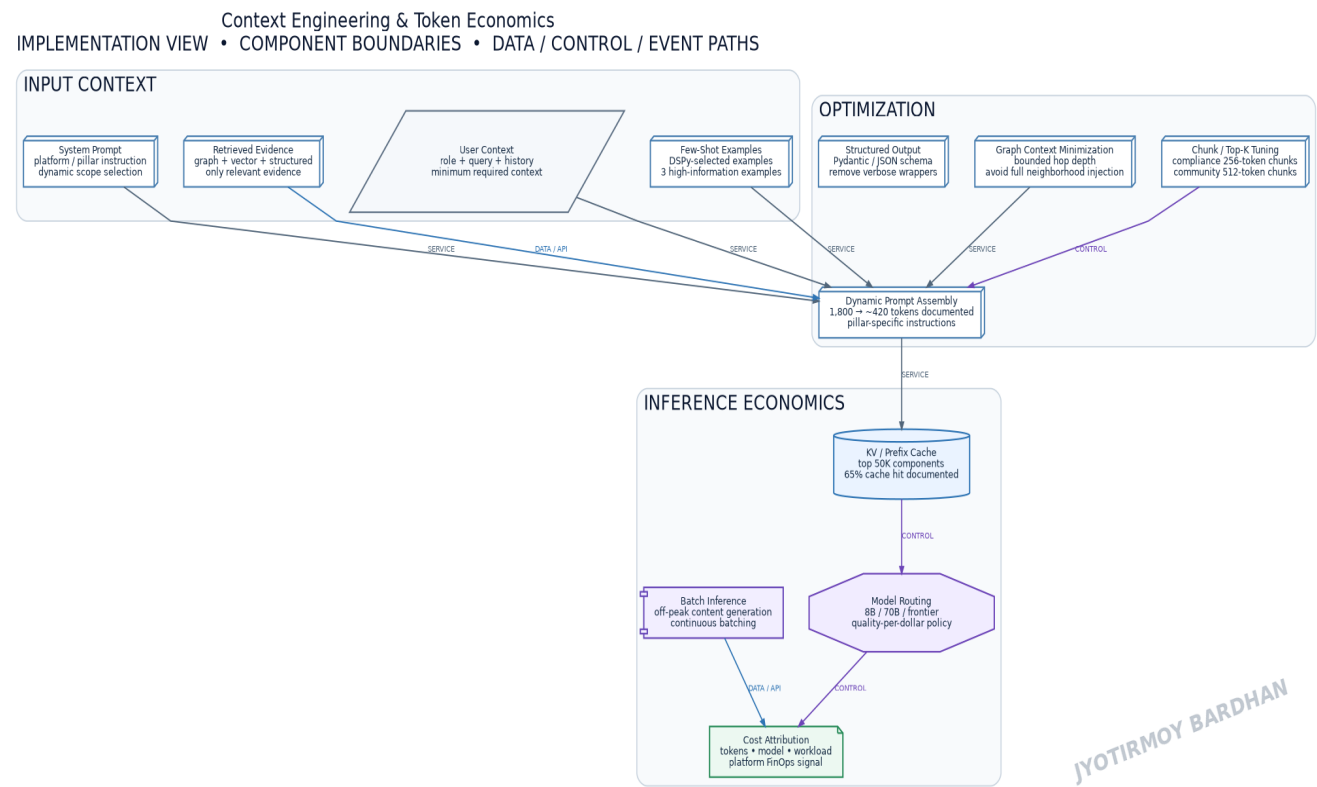
# 11. RAG / GraphRAG Retrieval Pipeline

IMPLEMENTATION VIEW • RAG / GraphRAG Retrieval Pipeline • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS



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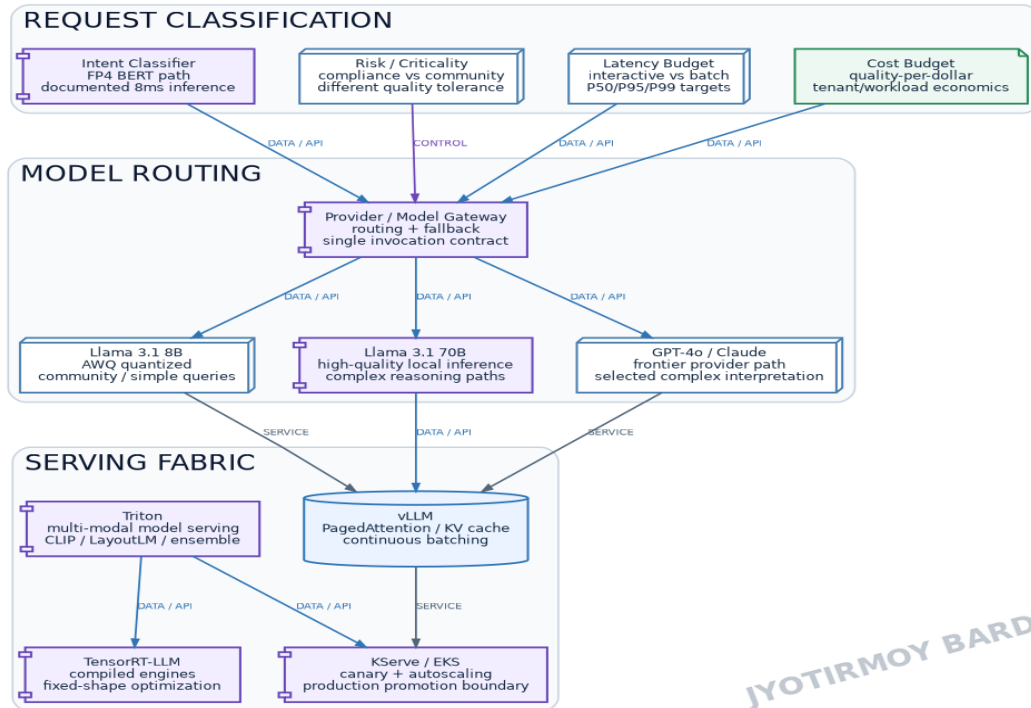
# 12. Context Engineering & Token Economics



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# 13. Multi-Model Routing & Inference

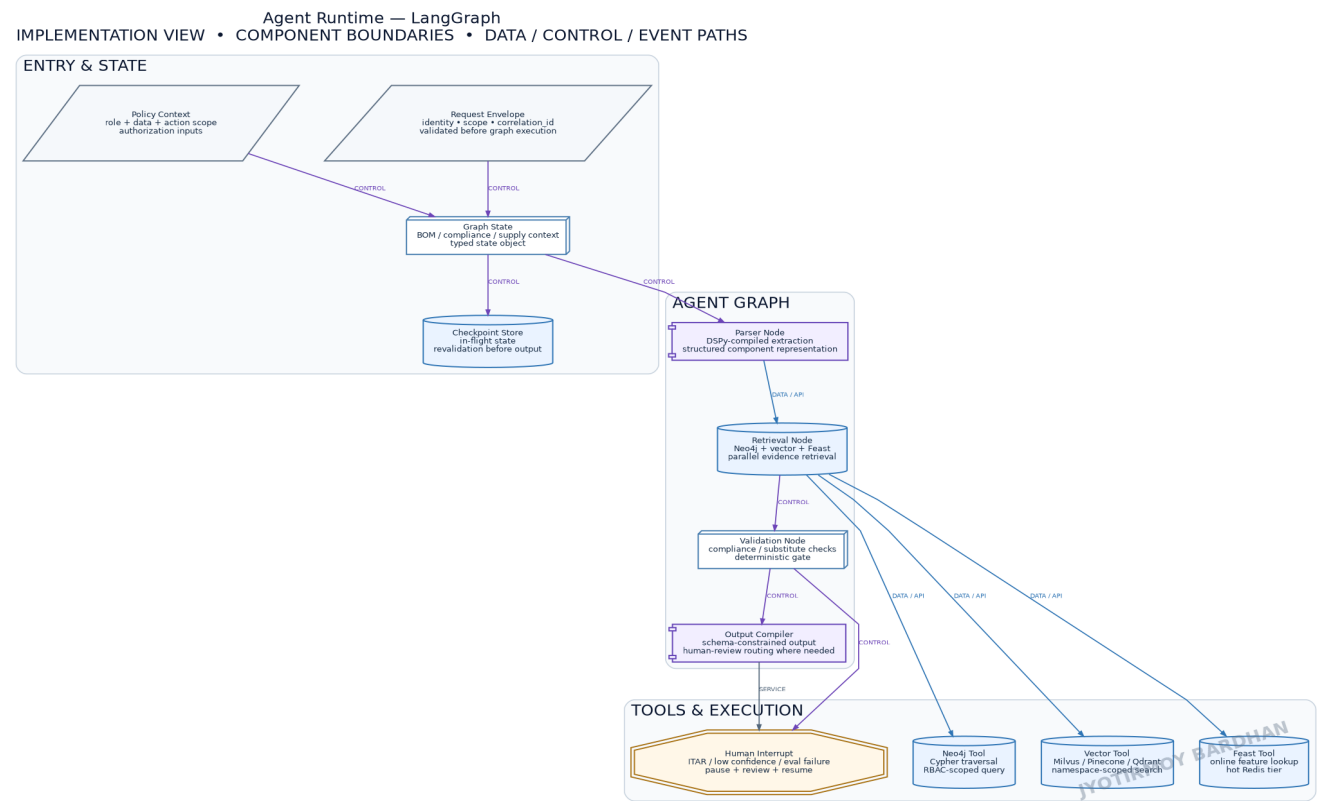
Multi-Model Routing & Inference  
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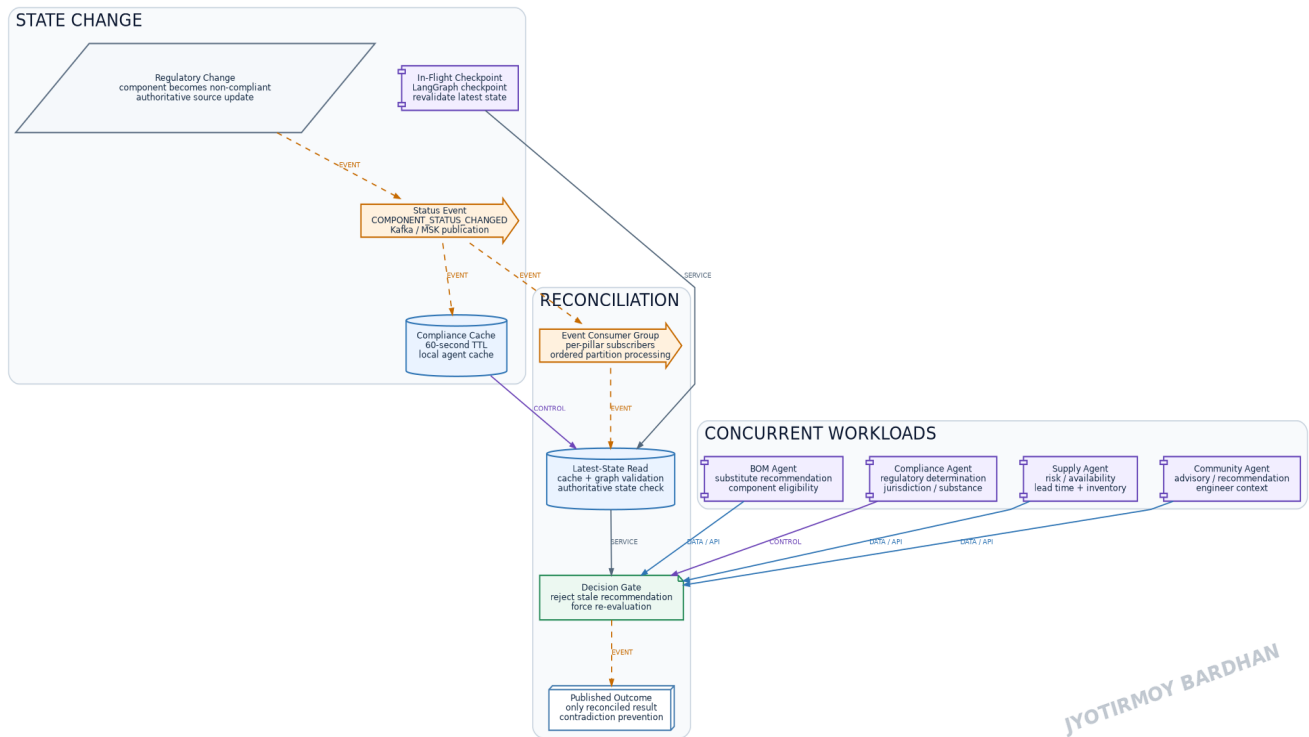
# 14. Agent Runtime — LangGraph



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# 15. State Consistency & Conflict Prevention

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 IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS

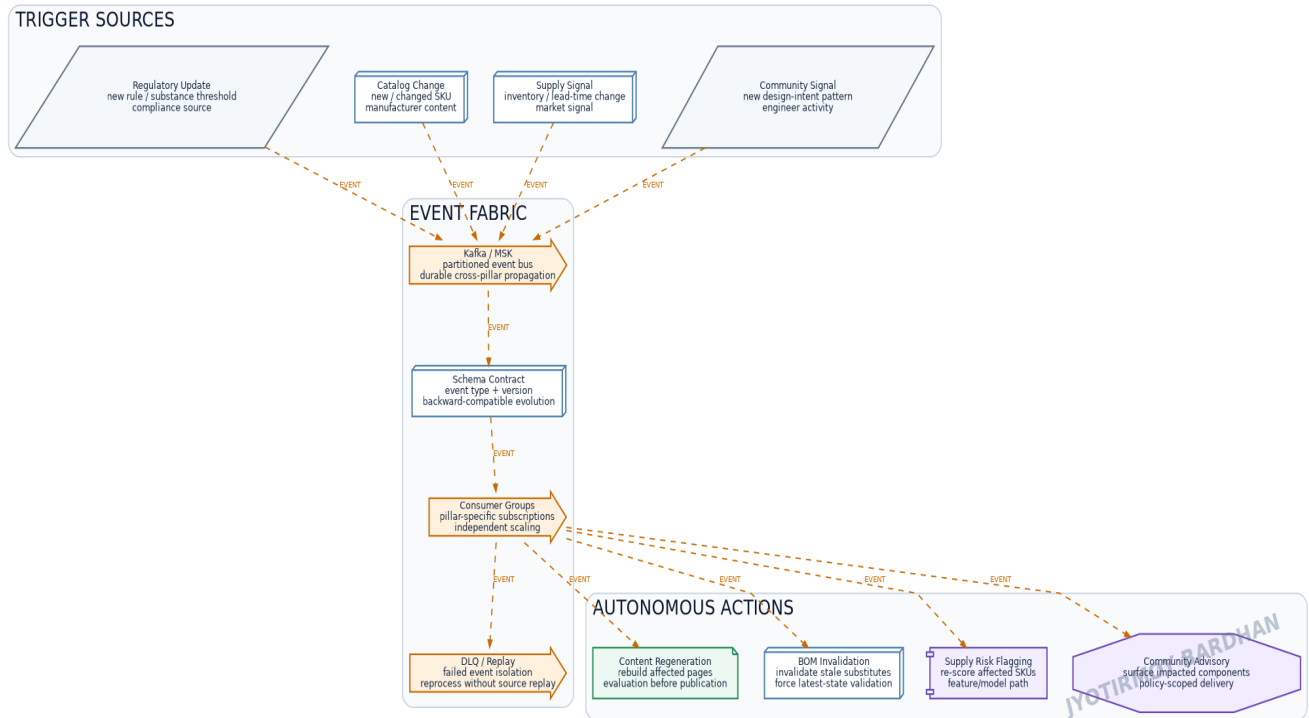


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# 16. Event-Driven Autonomous Workflow

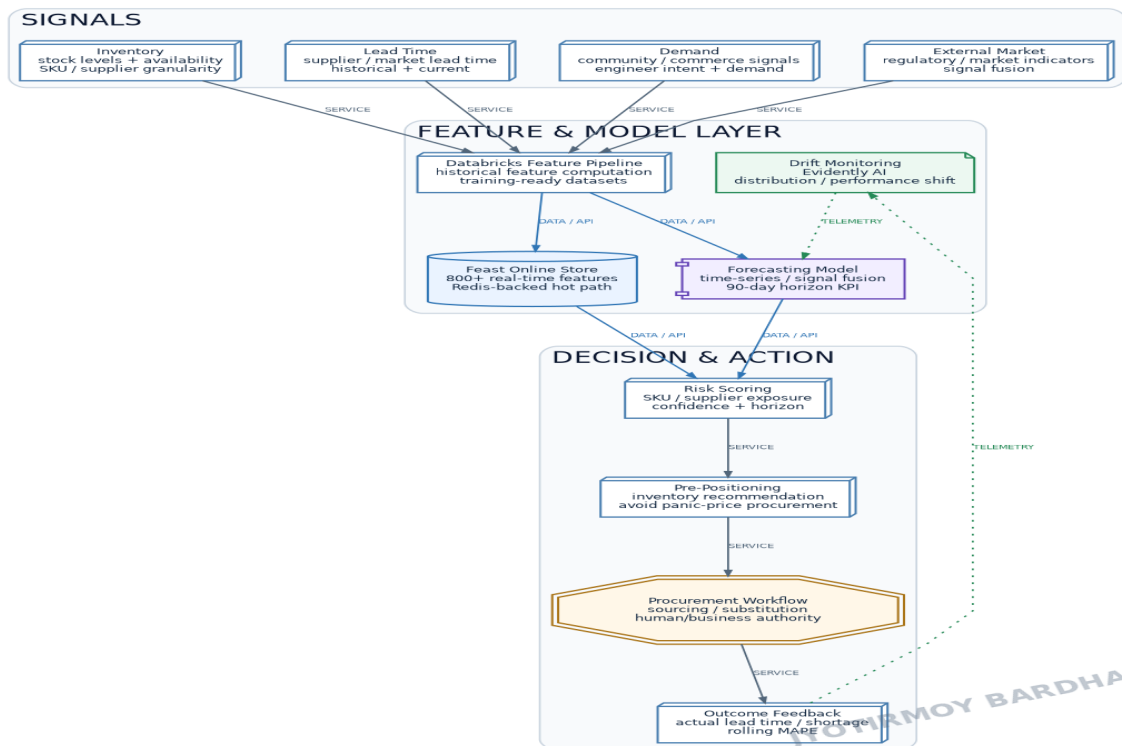
Event-Driven Autonomous Workflow  
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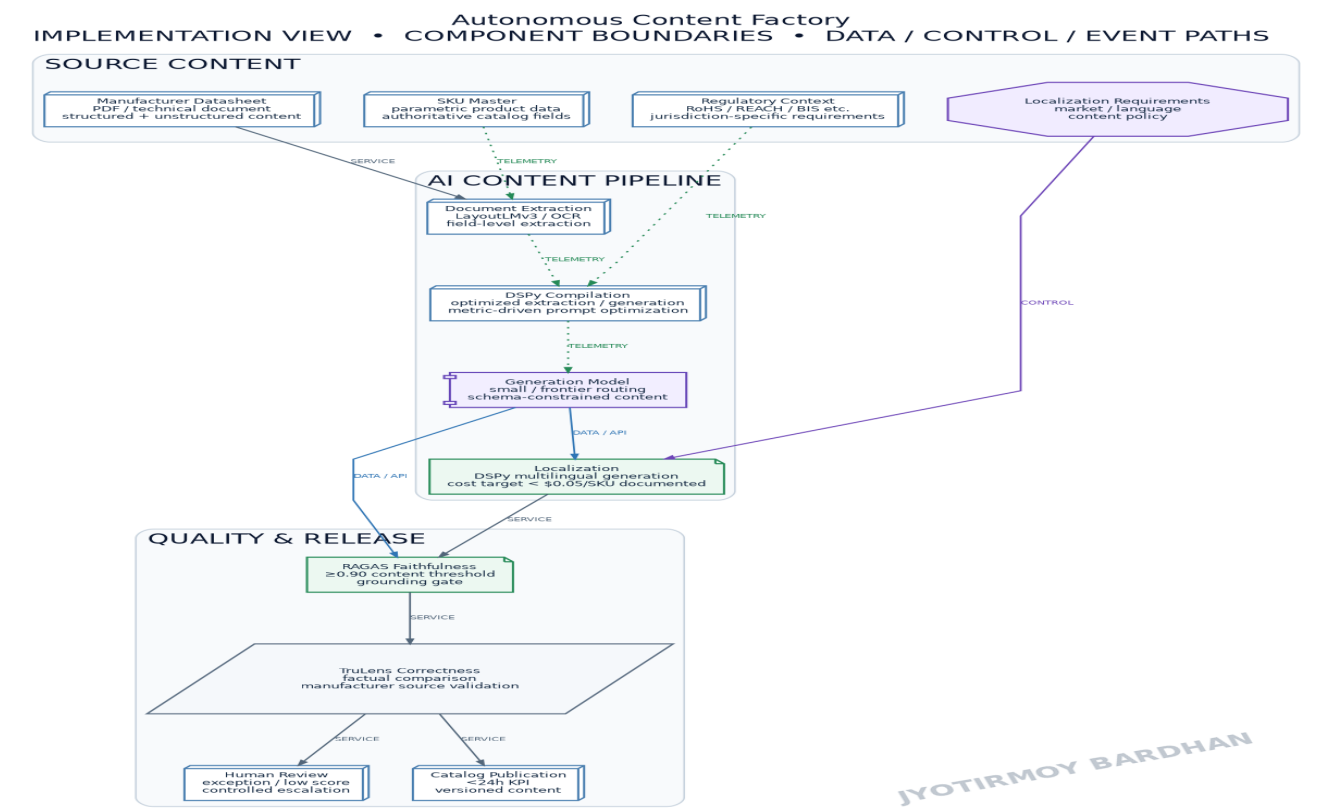
# 17. Supply Chain Resilience

Supply Chain Resilience  
IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS



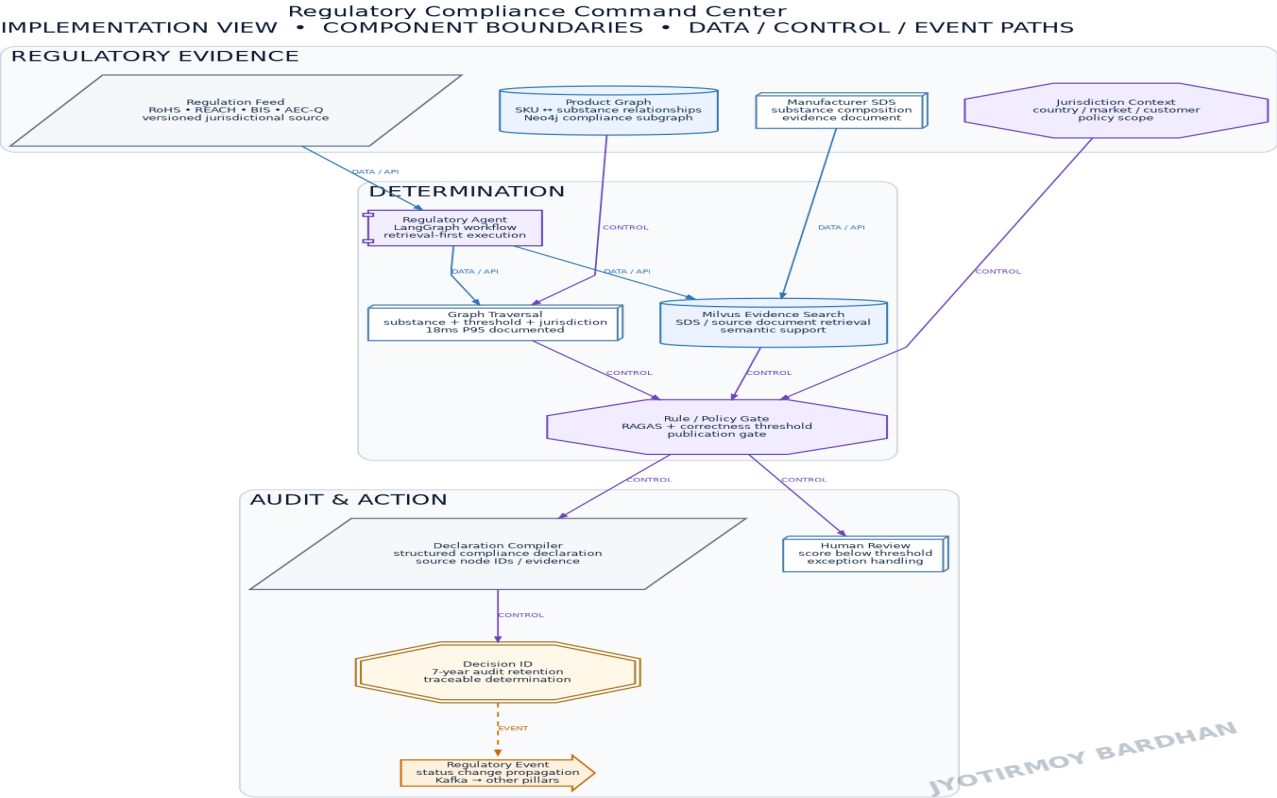
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# 18. Autonomous Content Factory



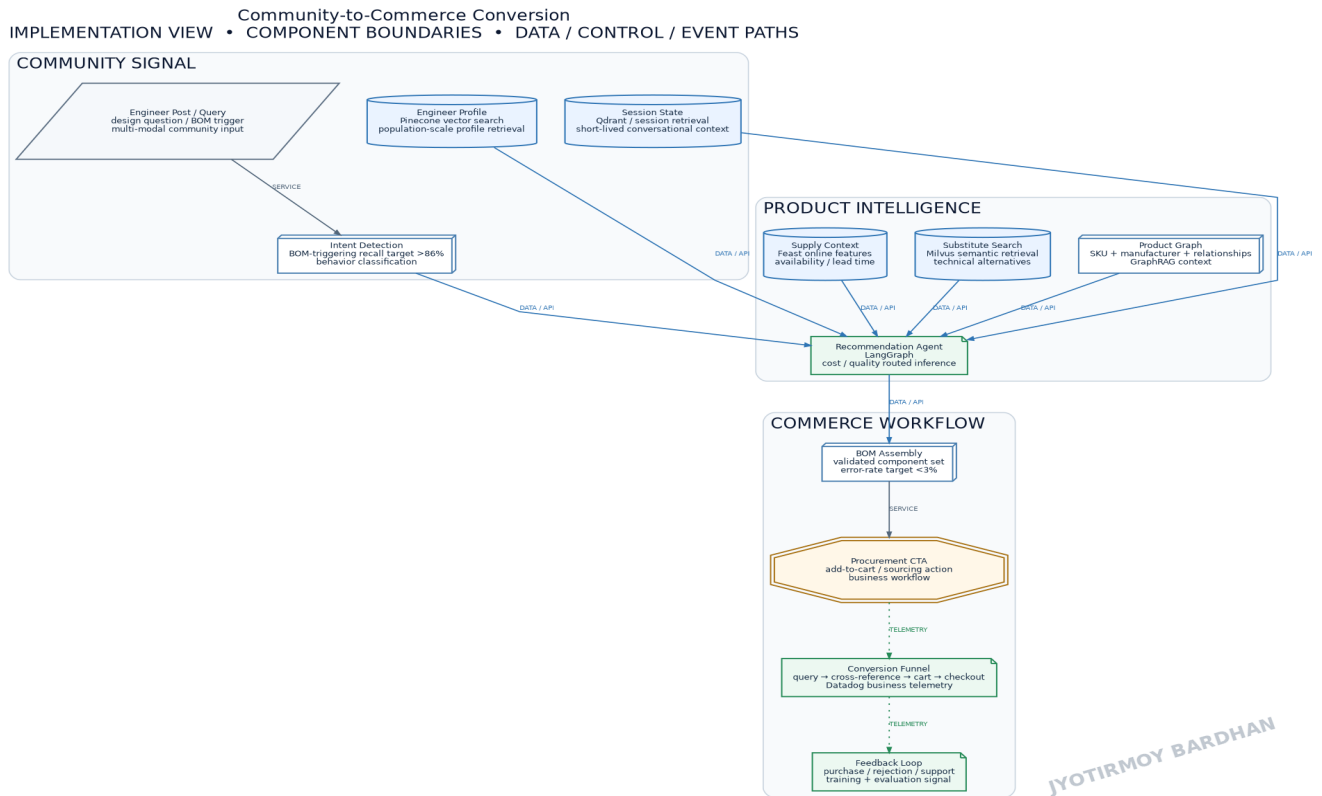
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# 19. Regulatory Compliance Command Center



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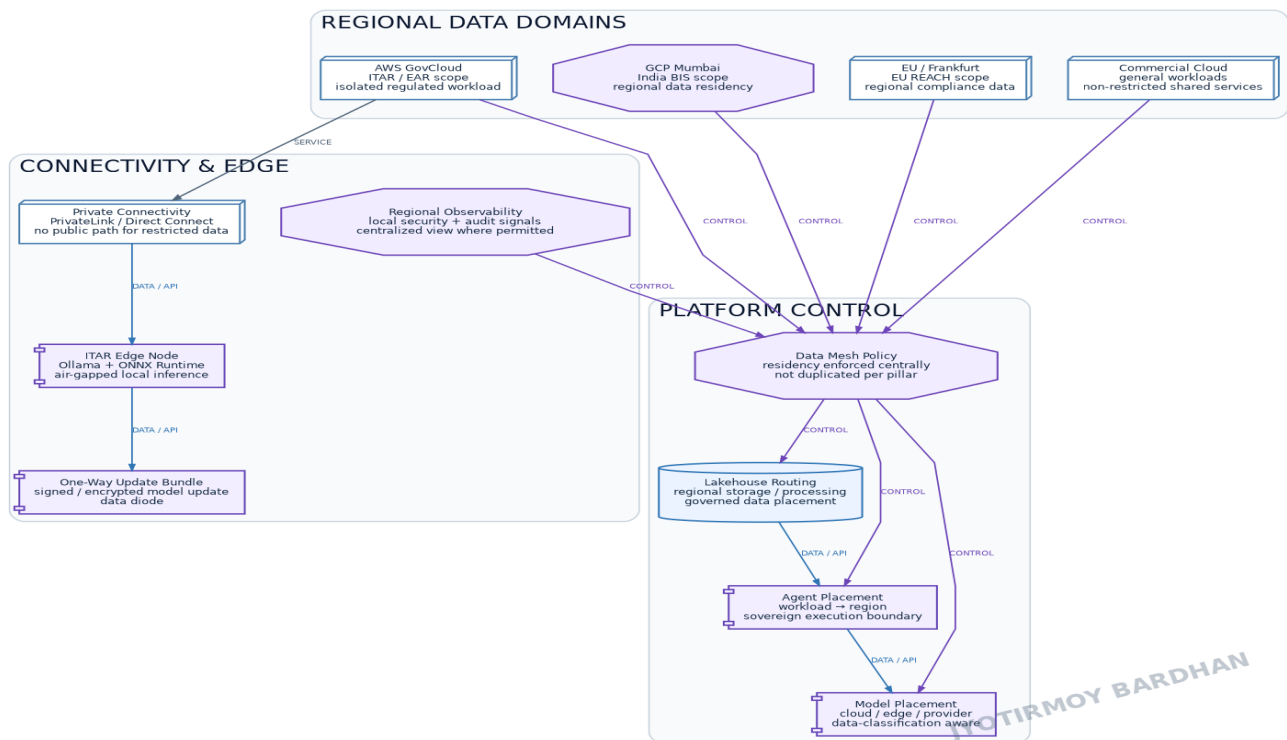
## 20. Community-to-Commerce Conversion



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## 21. Sovereign Multi-Cloud Deployment

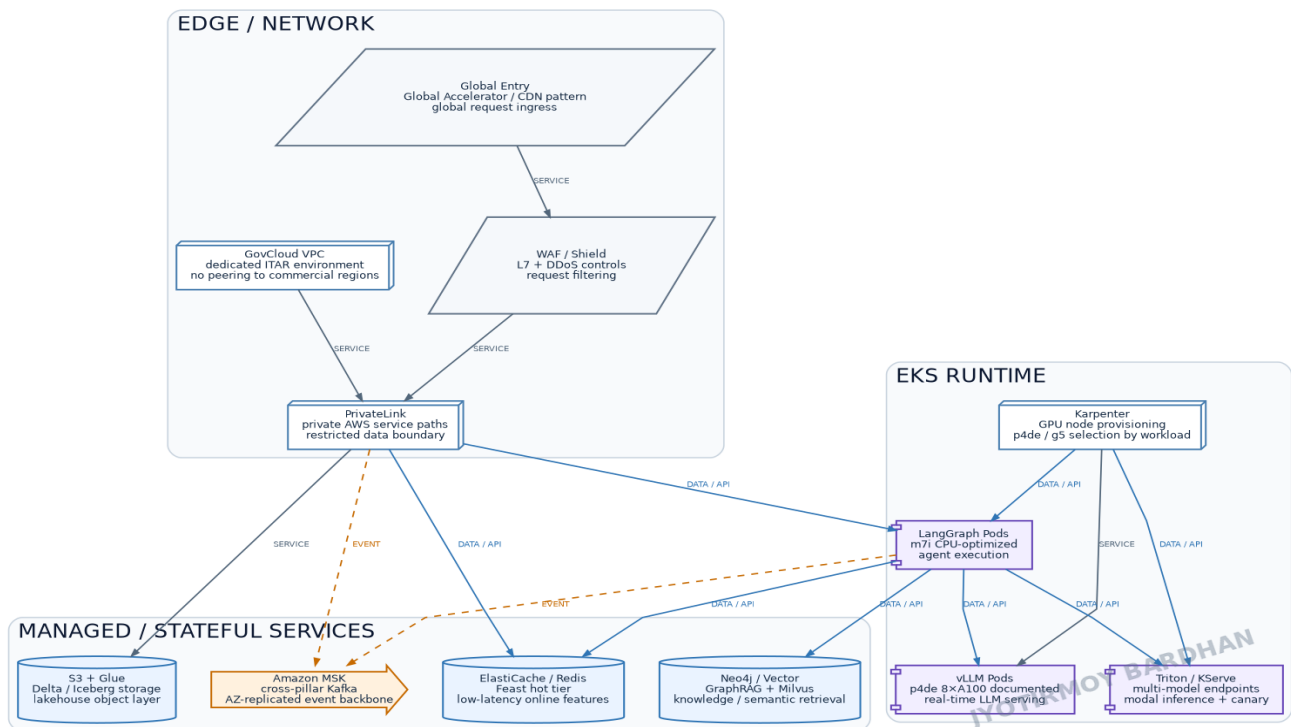
IMPLEMENTATION VIEW • SOVEREIGN Multi-Cloud Deployment • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS



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## 22. AWS Runtime Topology

AWS Runtime Topology  
IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS

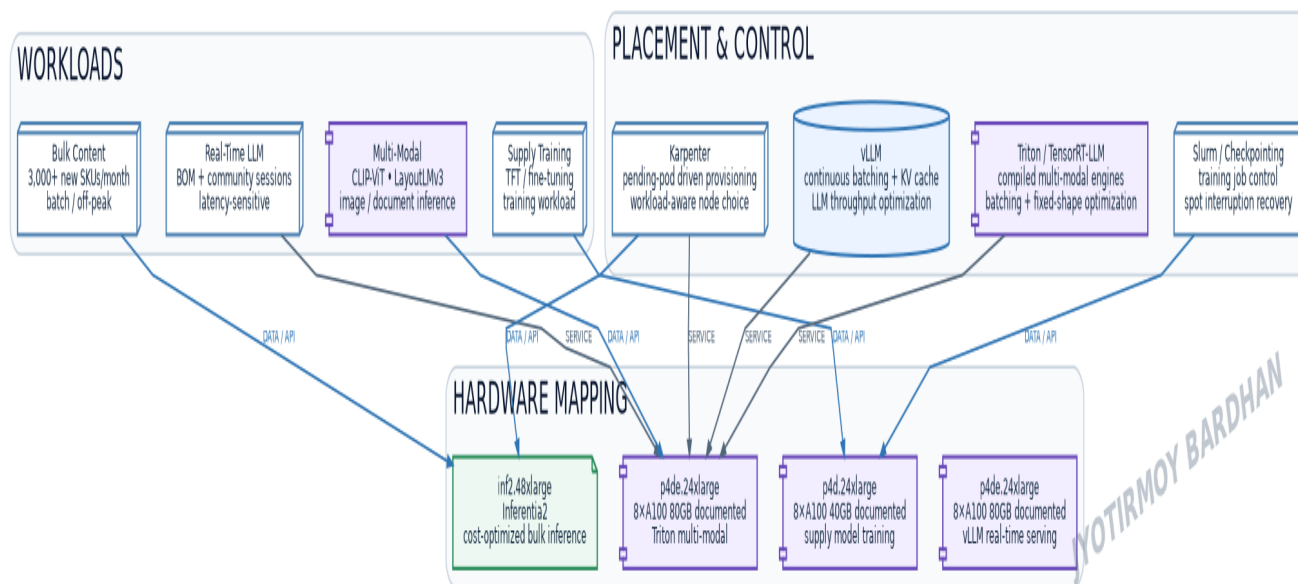


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## 23. GPU Serving & Hardware Placement

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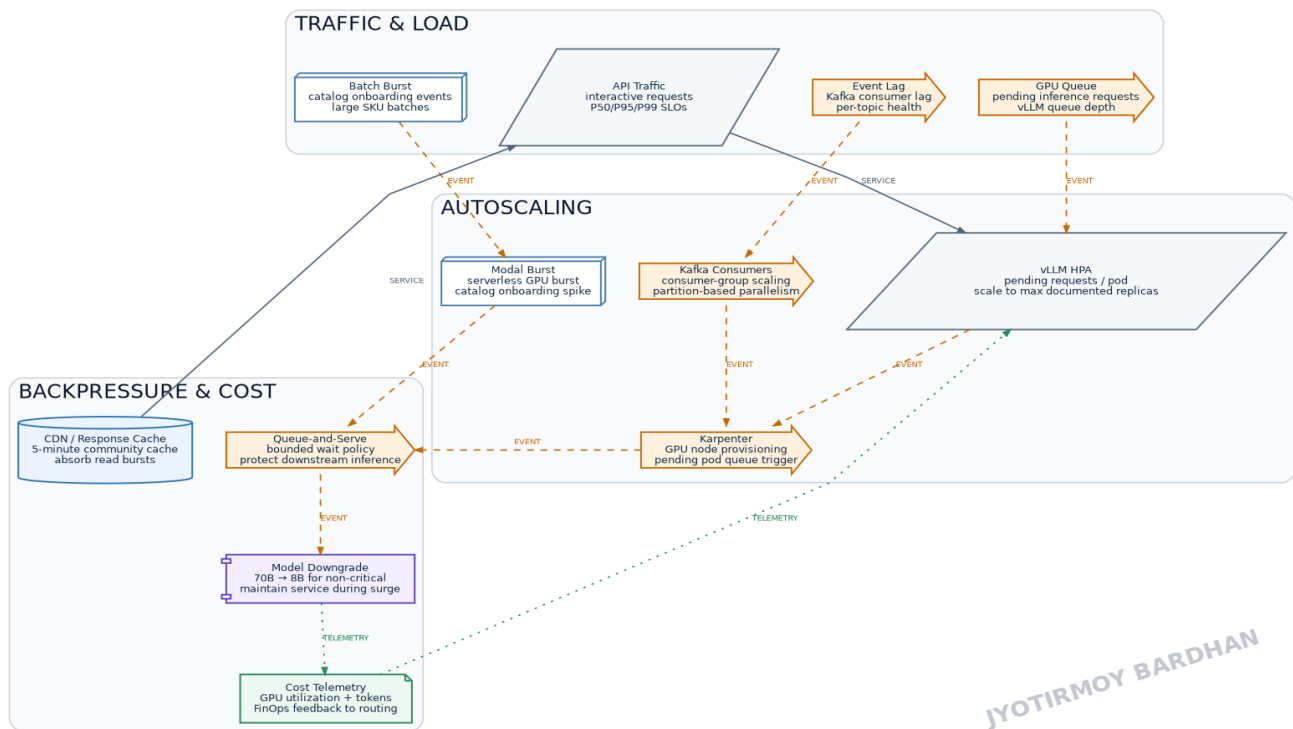
IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS



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## 24. Horizontal Scaling & Backpressure

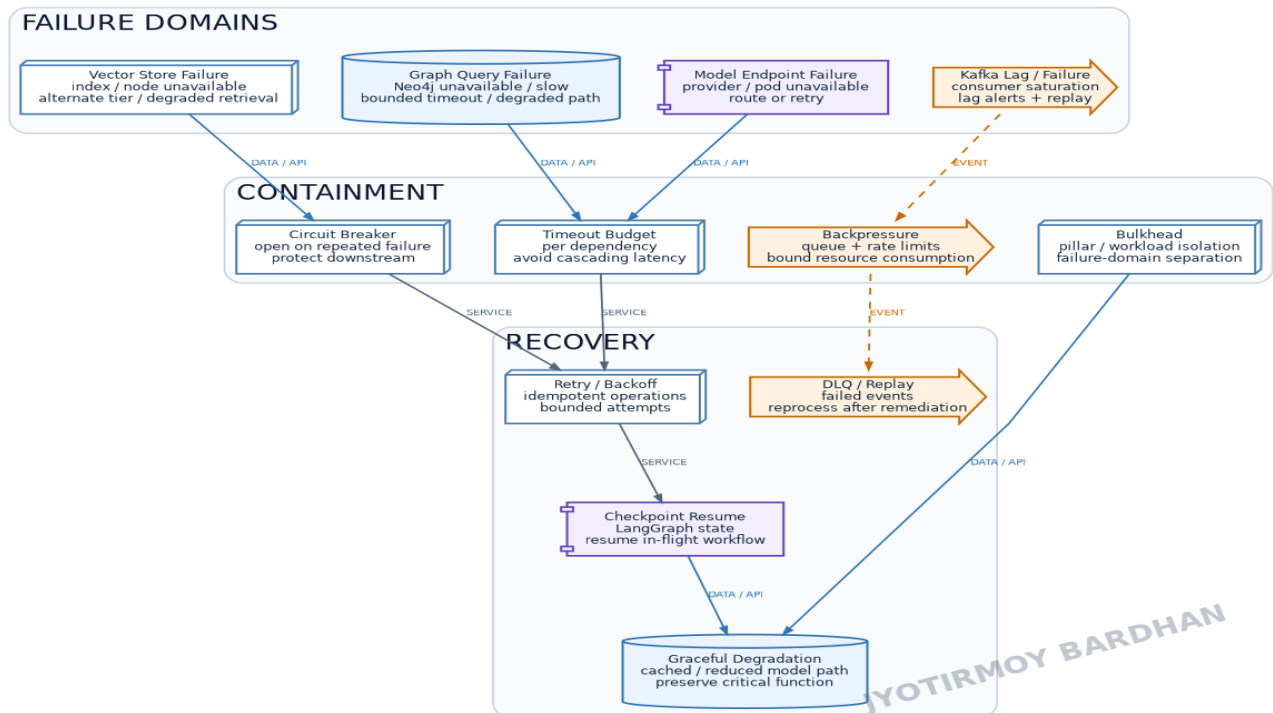
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## 25. Resilience & Failure Containment

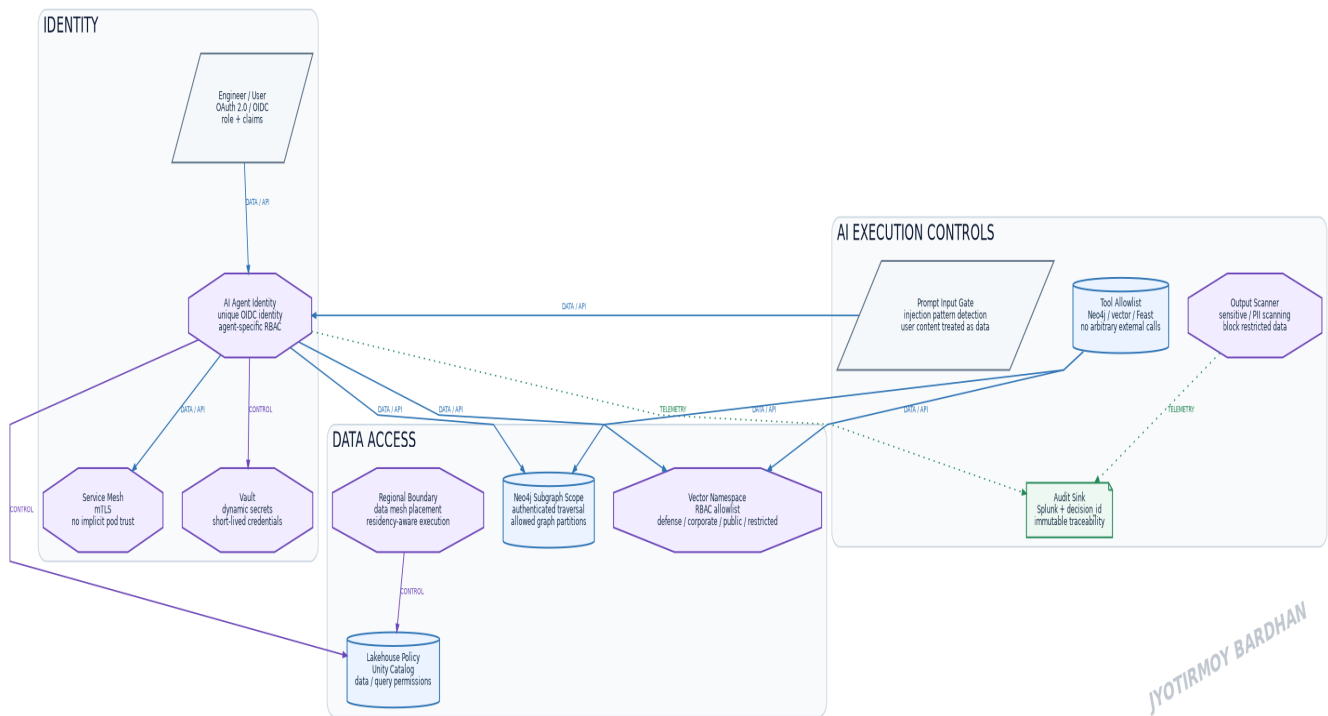
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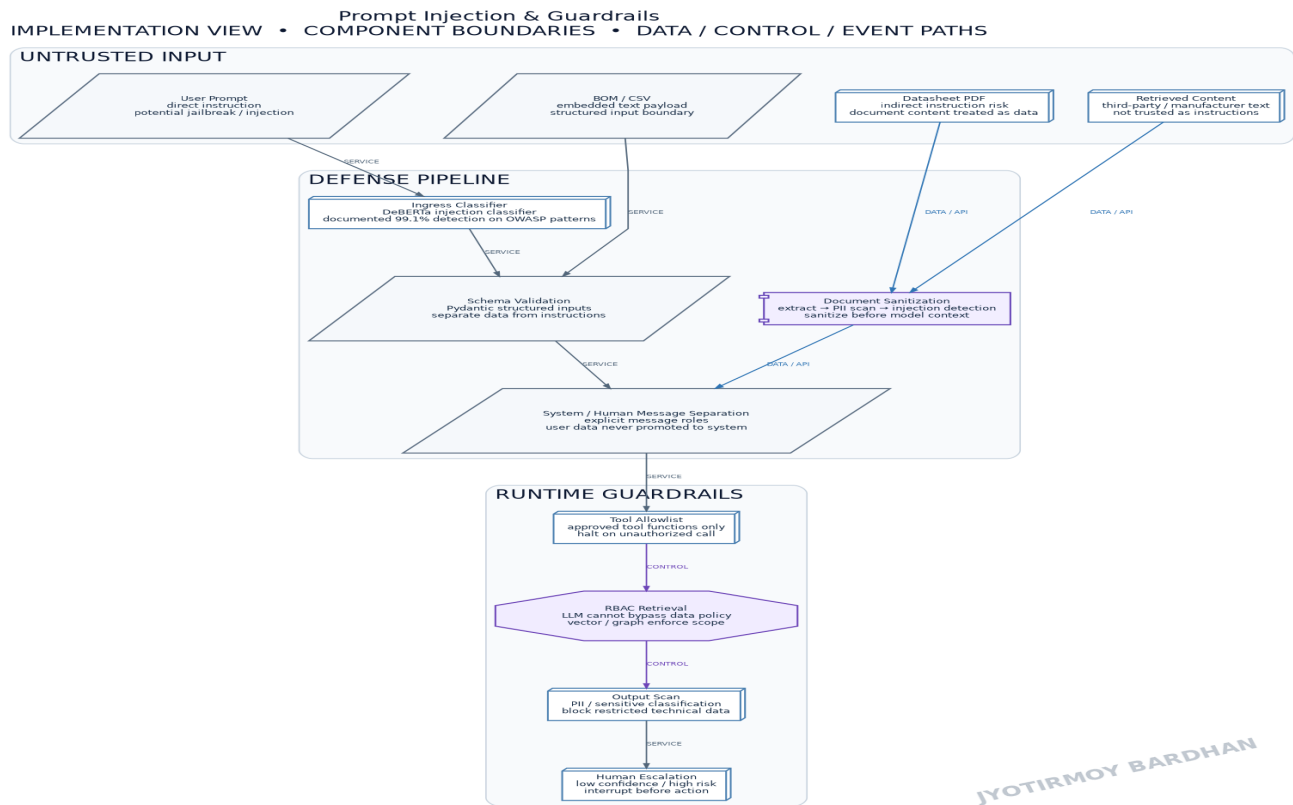
## 26. Zero-Trust Security Architecture

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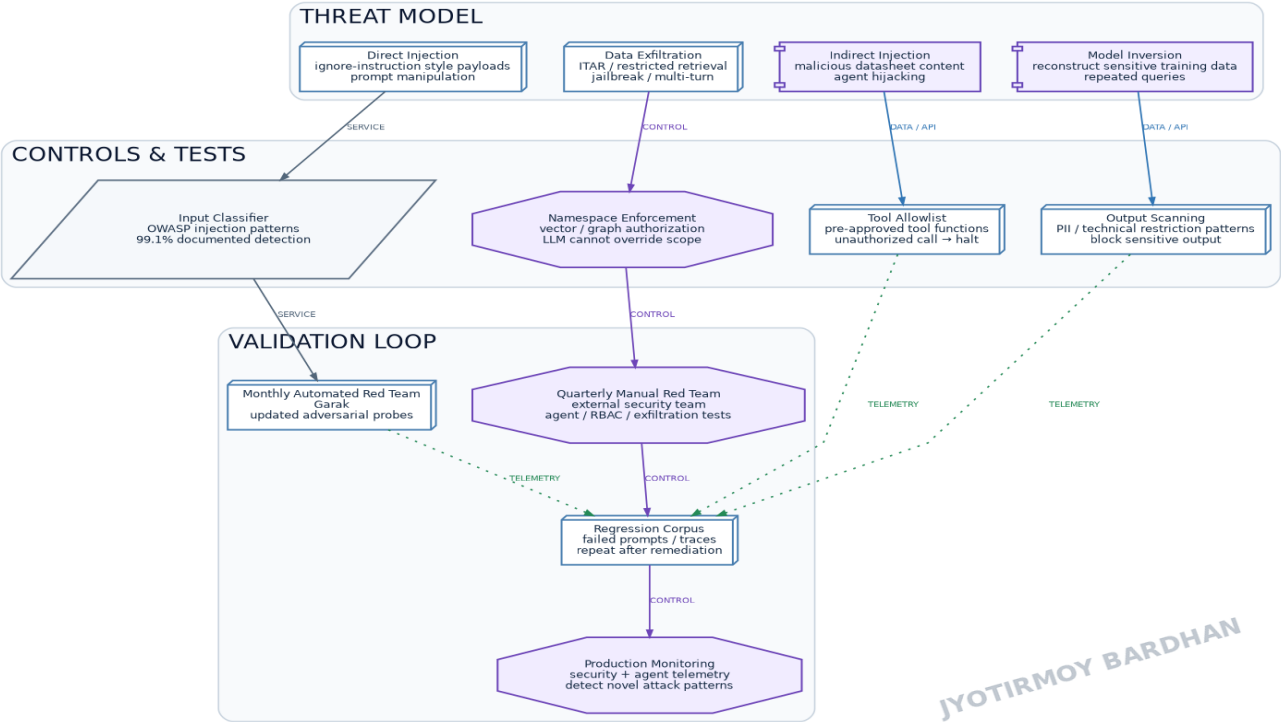
## 27. Prompt Injection & Guardrails



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# 28. Red-Team & Threat Model

Red-Team & Threat Model  
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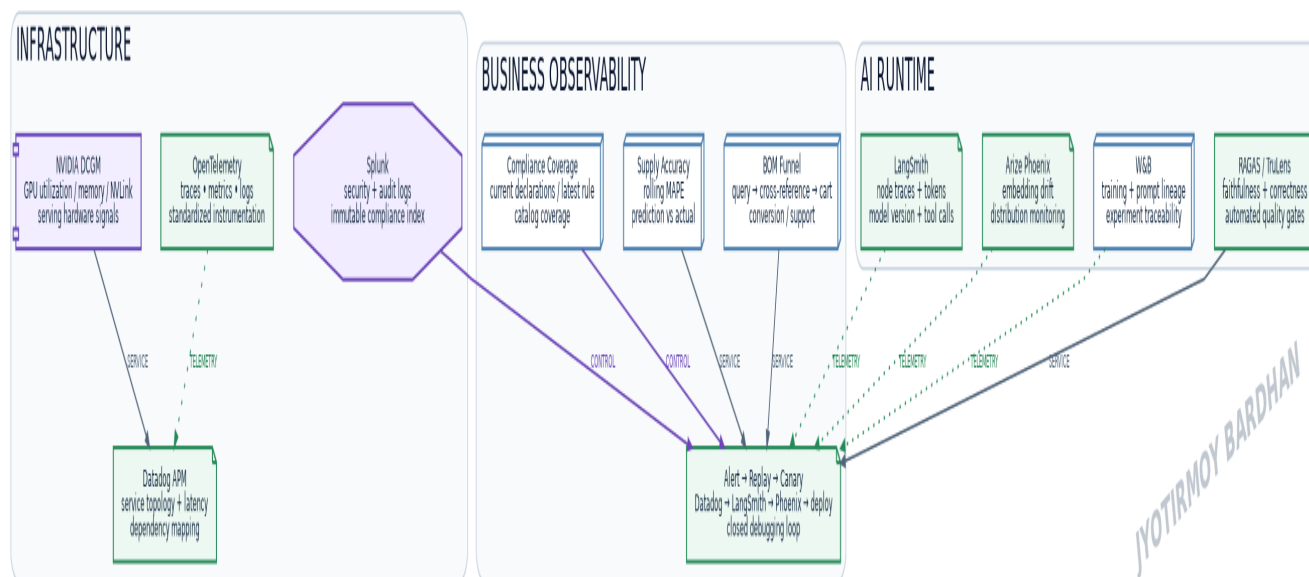


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## 29. Observability & Debugging

### Observability & Debugging

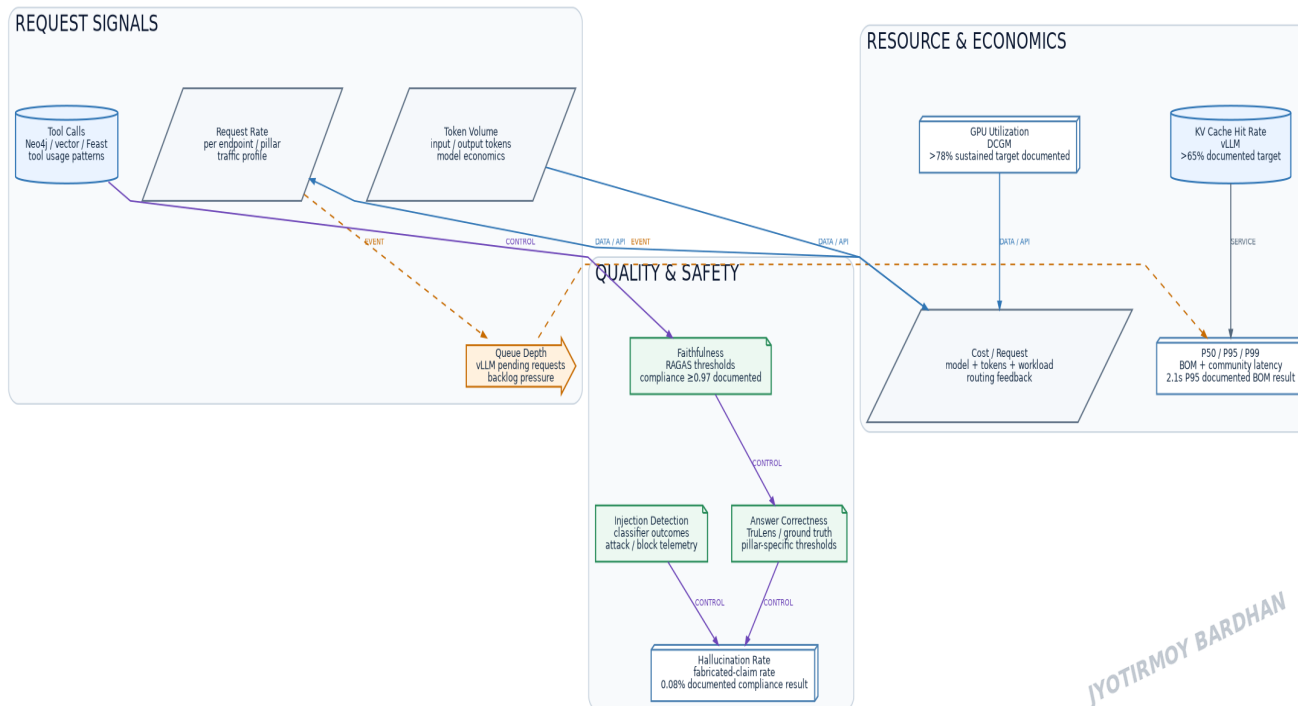
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# 30. Real-Time AI Runtime Monitoring

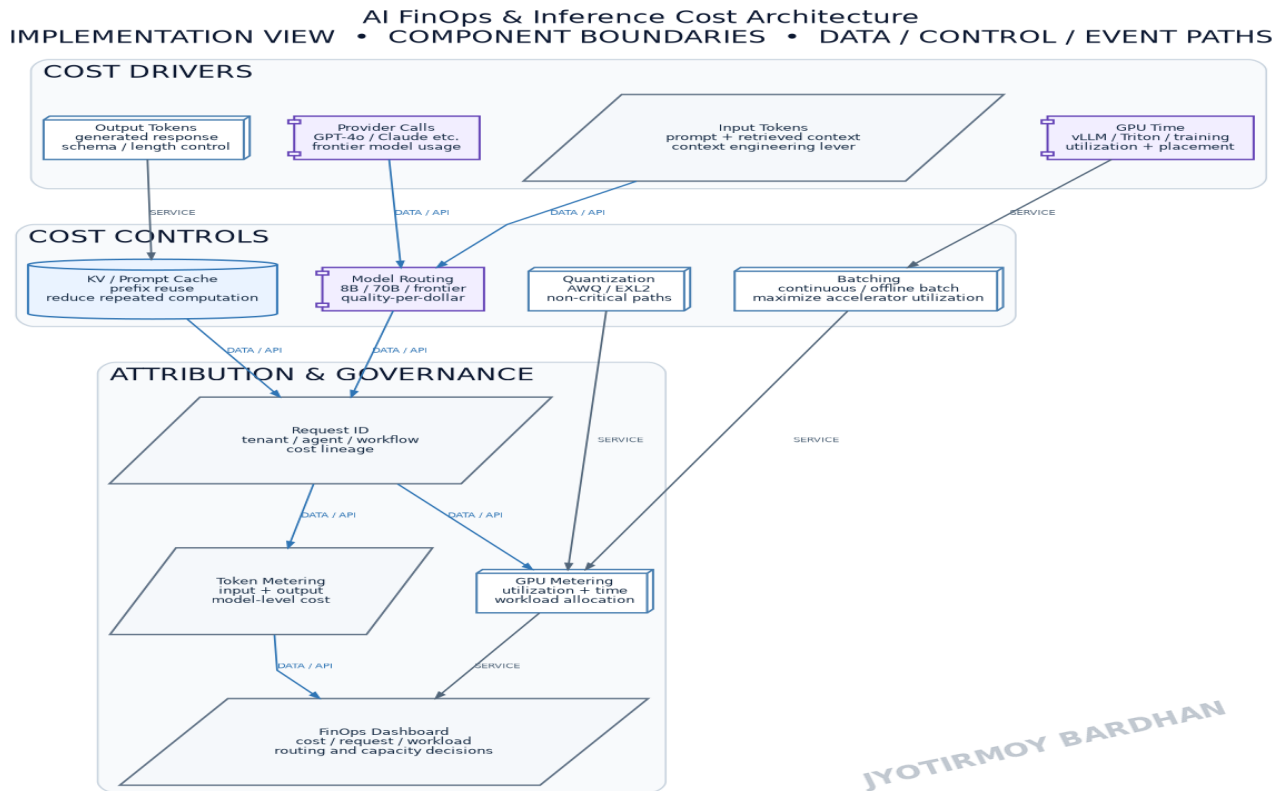
Real-Time AI Runtime Monitoring  
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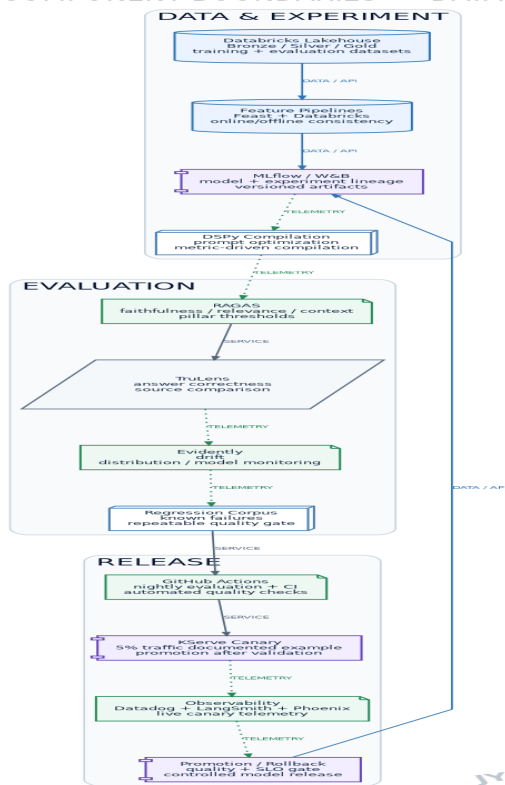
# 31. AI FinOps & Inference Cost Architecture



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## 32. MLOps / LLMOps Lifecycle

IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS

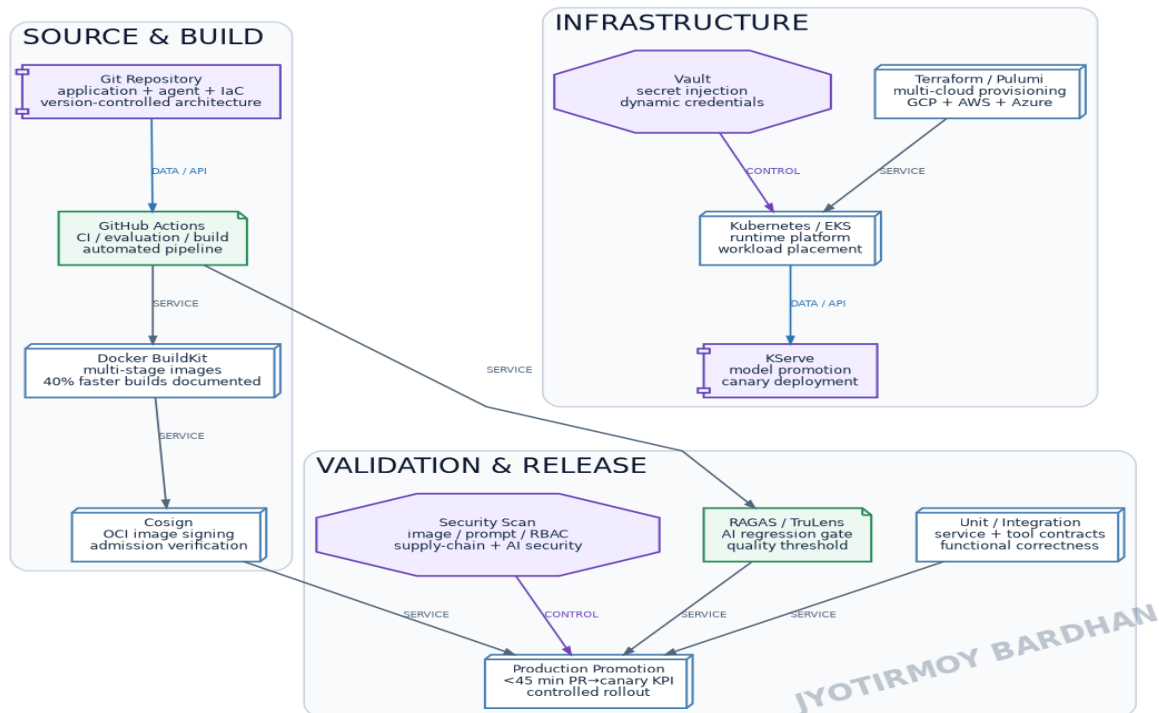


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### 33. CI/CD & Infrastructure Supply Chain

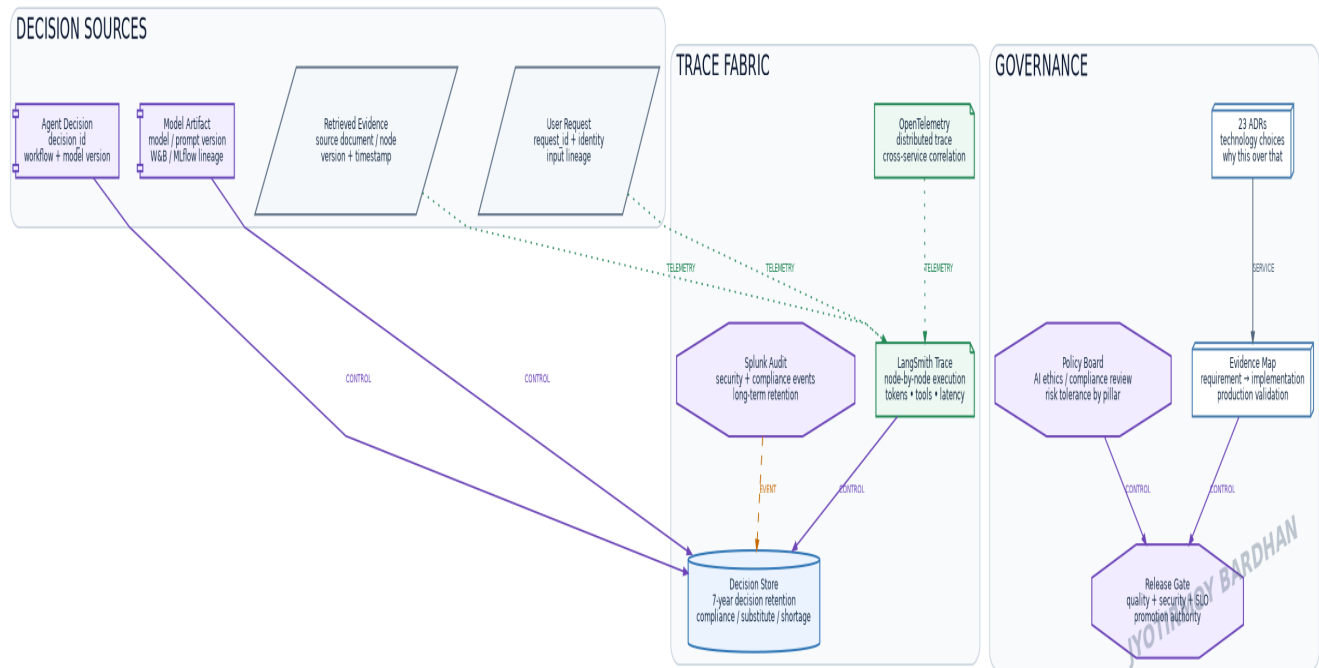
CI/CD & Infrastructure Supply Chain  
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# 34. Auditability, Traceability & Architecture Governance

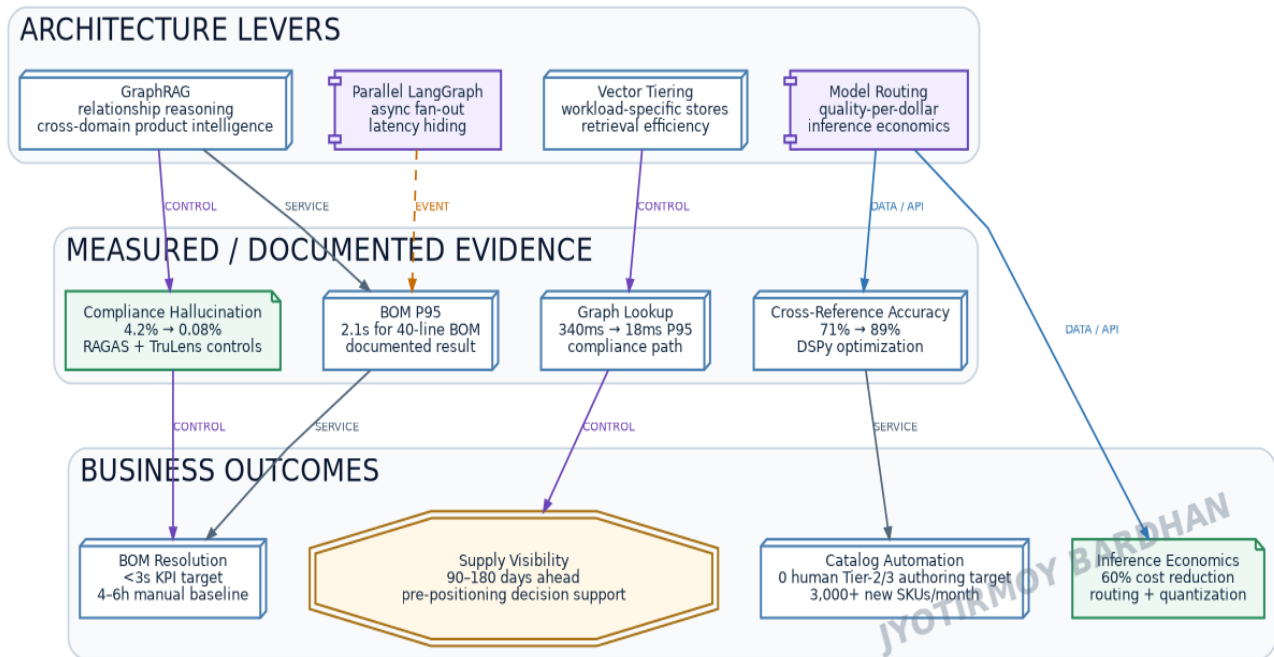
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## 35. Production Validation & Outcome Chain

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IMPLEMENTATION VIEW • COMPONENT BOUNDARIES • DATA / CONTROL / EVENT PATHS



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# Problem Framing

Five AI workstreams had different data models, retrieval methods and operational controls. The architecture reframed the problem as a shared intelligence substrate with explicit domain boundaries and governed cross-domain relationships.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Real-World Constraints

Documented constraints included heterogeneous product, compliance, supply and community data; sub-3-second BOM resolution; sovereign residency; air-gapped ITAR/EAR execution; cross-pillar state conflicts; and strict compliance hallucination tolerance.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Non-Functional Requirements

Performance, reliability, quality, security, sovereignty and operability were treated as architecture requirements. Documented examples include 2.1s P95 BOM resolution, 99.95% platform SLA, compliance faithfulness  $\geq 0.97$  and  $< 0.08\%$  compliance hallucination rate.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Integration Boundary

Source-system variability is contained at adapters and governed ingestion boundaries. Catalog, manufacturer, community, supply and regulatory inputs become reusable platform contracts rather than duplicated agent logic.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Canonical Product / Evidence Model

The knowledge fabric separates domain-owned subgraphs from centrally governed bridge relationships. Product, substance, supplier, design-intent and supply-risk relationships provide cross-pillar semantics.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Knowledge Graph Architecture

Neo4j is used for relationship-heavy reasoning and GraphRAG. Domain partitioning avoids an unbounded graph; teams own subgraph schemas while central architecture governs bridge relationships and query permissions.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Vector Retrieval Architecture

Milvus is documented for 1M+ SKU embeddings, Pinecone for engineer profiles and Qdrant for fast/session retrieval. Routing is workload-specific rather than based on a single universal vector store.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Retrieval Performance

Early sequential prototypes took 18–22 seconds. Parallel fan-out, graph prefetching, cache warming, connection pooling and workload-specific retrieval reduced the documented final 40-line BOM P95 to 2.1 seconds.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Context Engineering

Context was treated as a bounded resource through dynamic prompt selection, graph-context minimization, chunk/top-K tuning, structured outputs, few-shot reduction and prefix/KV-cache reuse.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Model Abstraction

The model gateway separates application intent from a specific model/provider. Routing selects local small models, larger local models or frontier providers according to quality, latency and cost constraints.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Agent Runtime

LangGraph provides stateful execution, conditional edges, checkpointing and human interruption. The runtime exposes allowlisted tools and revalidates compliance-sensitive state before delivery.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Cross-Pillar Consistency

COMPONENT\_STATUS\_CHANGED events propagate compliance state through Kafka/MSK. Local caches use a documented 60-second TTL, while in-flight checkpoints revalidate against latest state.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Event-Driven Architecture

The event fabric decouples regulatory, catalog, supply and community signals from downstream actions. Consumer groups scale independently and failed events can be isolated and replayed.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Supply Chain Intelligence

Inventory, lead time, demand and market signals feed feature pipelines and Feast online serving. Forecasting and risk scoring support pre-positioning decisions, with Evidently monitoring drift.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Content Automation

Datasheets and product data flow through extraction, DSPy compilation, generation, localization and evaluation. RAGAS/TruLens gates prevent low-quality content from automatic promotion.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Regulatory Intelligence

Regulatory determination is retrieval-first: regulation evidence, manufacturer SDS, product graph relationships and jurisdiction context are composed before generation. Decision IDs provide traceability.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Security Architecture

OIDC agent identities, Vault dynamic secrets, RBAC-scoped vector namespaces, Neo4j subgraph authorization, mTLS, private connectivity and sovereign placement enforce security outside the LLM.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Prompt Injection Defense

Direct and indirect injection are addressed through input classification, structured schemas, document sanitization, explicit message separation, tool allowlists, RBAC retrieval and output scanning.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Red Teaming

The documented program includes monthly automated Garak probes and quarterly manual adversarial testing. Findings feed a regression corpus and production monitoring.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Observability

OpenTelemetry, DCGM, Datadog and Splunk cover infrastructure. LangSmith, Phoenix and W&B; cover AI runtime and lineage. RAGAS/TruLens provide quality gates.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Real-Time Monitoring

Runtime monitoring separates request rate, queue depth, token volume, tool calls, latency, faithfulness, correctness, hallucination, injection detection, GPU utilization, cache hit rate and cost/request.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# AI FinOps

Inference cost is decomposed into tokens, GPU time and provider calls. Routing, caching, quantization and batching reduce avoidable spend while request/model/workload attribution closes the feedback loop.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Scalability

Karpenter provisions GPU nodes, vLLM scales serving pods, Kafka consumer groups scale by partition demand and burst capacity absorbs catalog events. Queueing and controlled model downgrades protect critical paths.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Resilience

Failure containment uses dependency timeouts, circuit breakers, bulkheads, backpressure, bounded retry, DLQ/replay and checkpoint resume. Graceful degradation preserves useful service.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Cloud Deployment

The documented topology uses AWS, GCP and Azure with regional data-residency controls. AWS EKS is the primary runtime baseline, with GovCloud isolation and an air-gapped edge path for ITAR/EAR workloads.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# MLOps / LLMOps

Databricks, MLflow/W&B;, DSPy, RAGAS, TruLens, Evidently and GitHub Actions form the model/prompt lifecycle. KServe canary promotion couples release with live quality and SLO telemetry.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# CI/CD, IaC & Auditability

GitHub Actions, Docker BuildKit, Cosign and Terraform/Pulumi provide reproducible delivery. LangSmith, OTEL, Splunk and decision IDs preserve runtime traceability; 23 documented ADRs capture major technology decisions.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.

# Principal Architect Contribution & Production Validation

The documented role included end-to-end architecture vision, graph schema and bridge standards, sovereign data residency design, vector-tier strategy, model routing, ADR ownership, governance, C-suite alignment, budget ownership and hands-on validation of graph queries, DSPy, KV-cache, LangGraph and evaluation pipelines.

**Architecture trace:** requirement → constraint → decision → implementation → telemetry → documented outcome.